BUFERS AND pH

LEARNING OBJECTIVES

At the end of lecture the student should be able to;

- Define pH
- Discuss the hydrogen ion production in the body
- Define buffers
- Explain the mechanism of action of buffers
- Justify that bicarbonate buffer is the major buffer of the body.
- Discuss the clinical states of disturbed pH

DEFINATION

- pH
  - Negative log of Hydrogen ion concentration.
- pOH
  - Negative log of hydroxyl ion concentration
- pK
  - Negative log of dissociation constant
- pK_a
  - Negative log of weak acid
- H-H equation
  - $pH=pK_a$ + log base /acid

Definitions

- pH= negative log of Hydrogen ion concentration.
- pKa= negative log of dissociation constant of the acid
- A= conjugate base
- HA= weak acid

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

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table 2.5 Hydrogen Ion Concentrations and pH

Grams of H^+ per Liter	pH	
0.00000000000001	14	↑ Increasingly basic
0.0000000000001	13	
0.000000000001	12	
0.00000000001	11	
0.0000000001	10	
0.000000001	9	
0.00000001	8	Neutral–neither acidic nor basic
0.0000001	7	
0.000001	6	
0.00001	5	
0.0001	4	
0.001	3	Increasingly acidic
0.01	2	
0.1	1	↓
1.0	0	

THE BODY AND pH

- Homeostasis of pH is tightly controlled
- Extracellular fluid = 7.4
- Blood = 7.35 – 7.45
- < 6.8 or > 8.0 death occurs
- Acidosis (acidemia) below 7.35
- Alkalosis (alkalemia) above 7.45

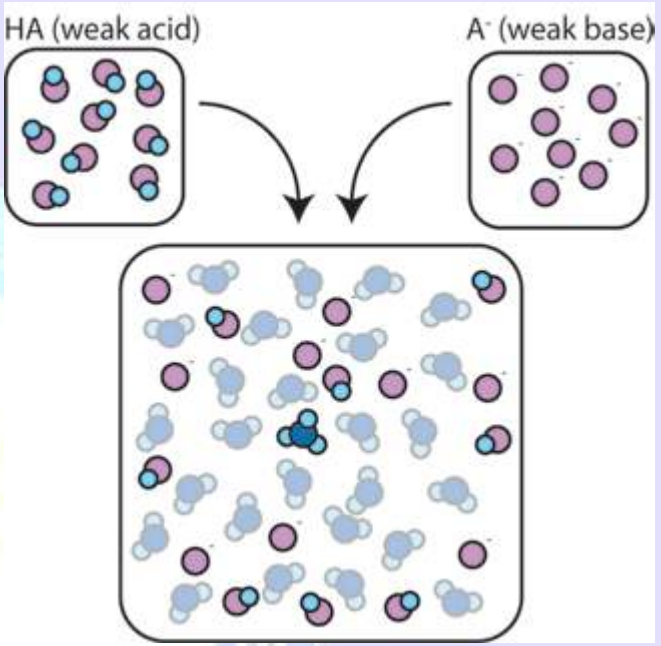
Small Changes In pH Can Produce Major Disturbances

- Most enzymes function only with narrow pH ranges
- Acid-base balance can also affect electrolytes (Na^+, K^+, Cl^-)
- Can also affect hormones

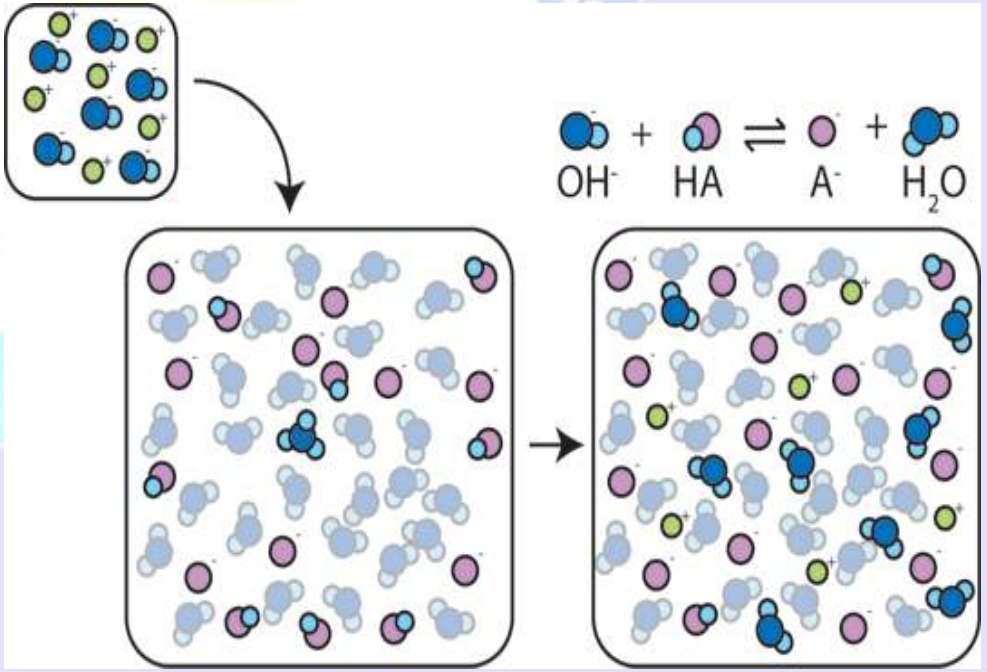
Sources of H^+ in the body

- Metabolism of non-carbon groups
  - Oxidation of sulphur $2SH \longrightarrow 2S + 2H^+$
  - Metabolism of Urea 2NH3 urea + $2H^+$
- Incomplete metabolism
  - Anaerobic glucose metabolism
  - Fatty acid metabolism
- Respiration
  - At lung level CO_2
  - Oxidative phosphorylation

Control of Acids
A. Buffer systems



- Buffers
  - Tend
  - Resist
  - The change in pH
  - When acid or alkali is added
  - In small amounts
- Buffer pair (acid/conjugate base or salt). When
  - Strong acid is added
    - Conjugate base + Strong acid →Salt
  - Base is added
    - Acid + Strong base →Salt + water



BUFFERING CAPACITY

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

- When an acid is exactly half-neutralized, [A−] =[HA].
- Under these conditions,
- at half-neutralization, pH = p*Ka.*
- Equal amount of acid or conjugate base is available for neutralization
- Therefore buffering capacity depends upon the pKa i.e. nearer to pH
- Concentration of the weak acid/salt pair

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

$$= pK_a + \log \frac{1}{1}$$

$$= pK_a + 0$$

$$pH = pK_a$$

DISTRIBUTION OF BUFFERS

- Extra-cellular (bicarbonate, proteins)
- Blood Buffers (bicarbonate, plasma proteins, hemoglobin, phosphate)
- Intra-cellular (phosphate, proteins, bicarbonate)
- Renal (phosphate, ammonia, bicarbonate)
- Bone (calcium carbonate, calcium phosphate)

The most important buffer

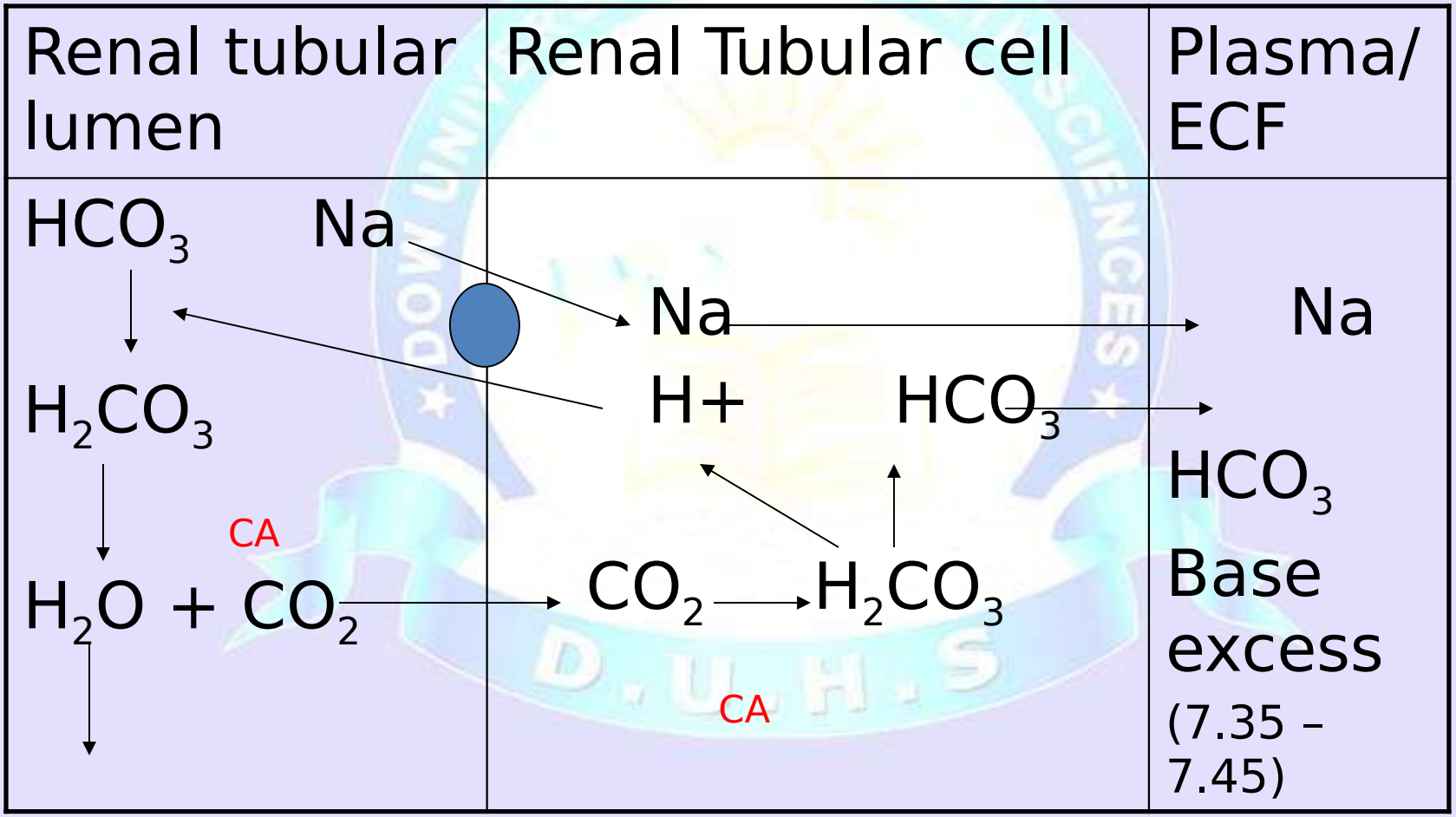
- Carbonic Anhydrase

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

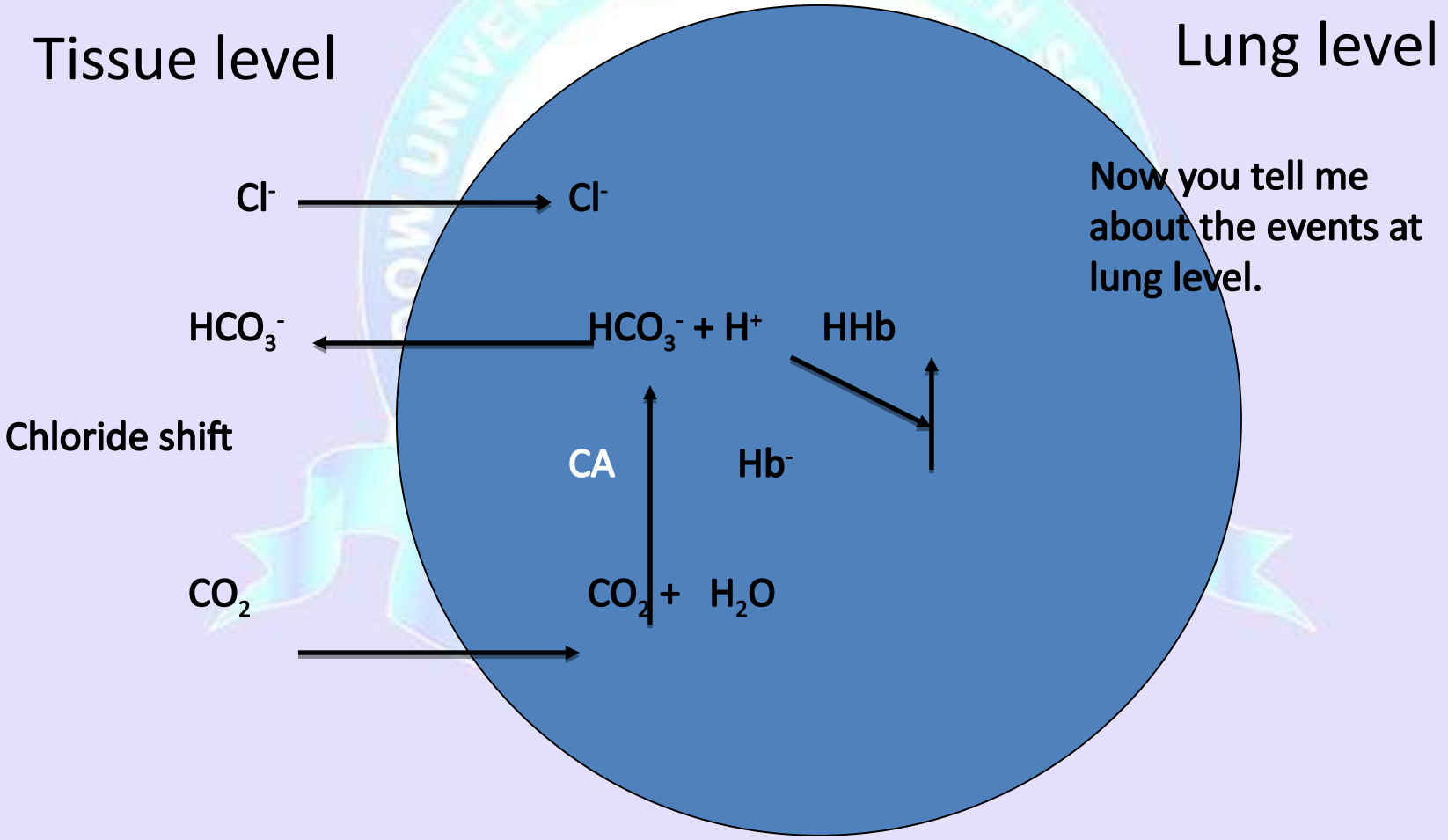
Sites:

RBC, lung epithelium, stomach, renal tissue, intestinal tissue

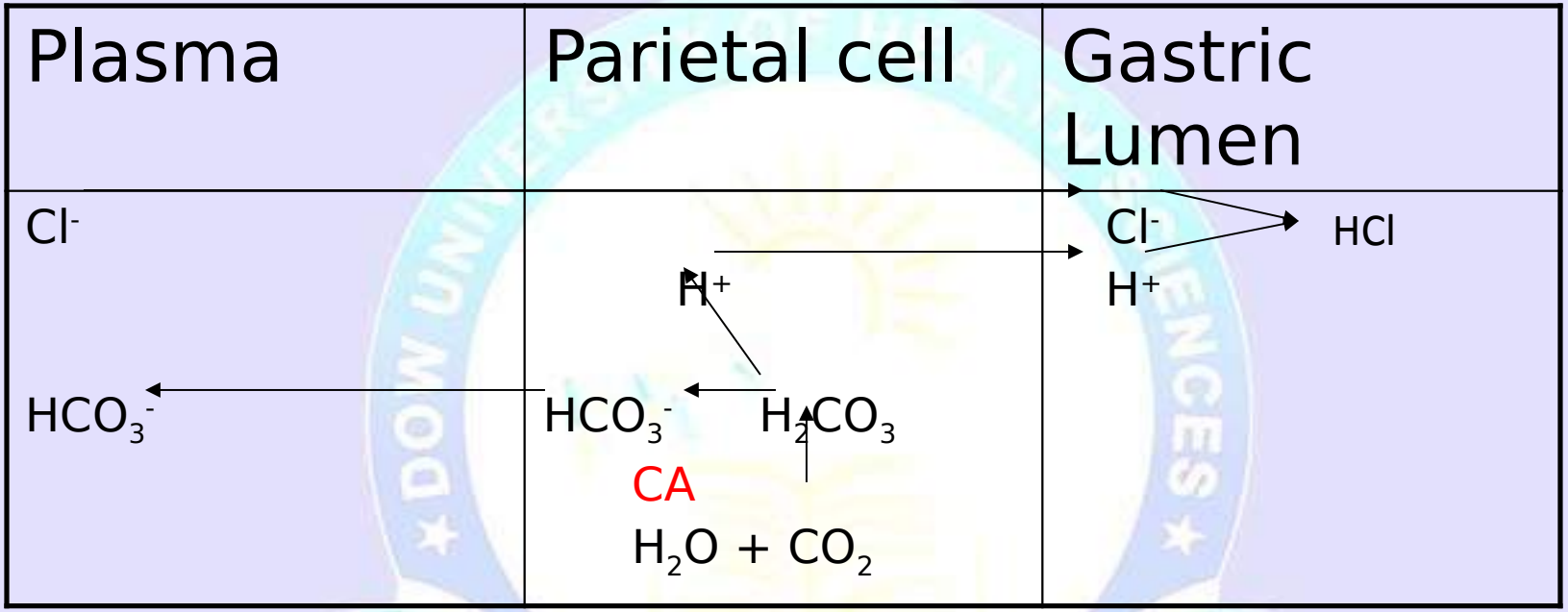
Bicarbonate buffers maintains alkalinity of blood



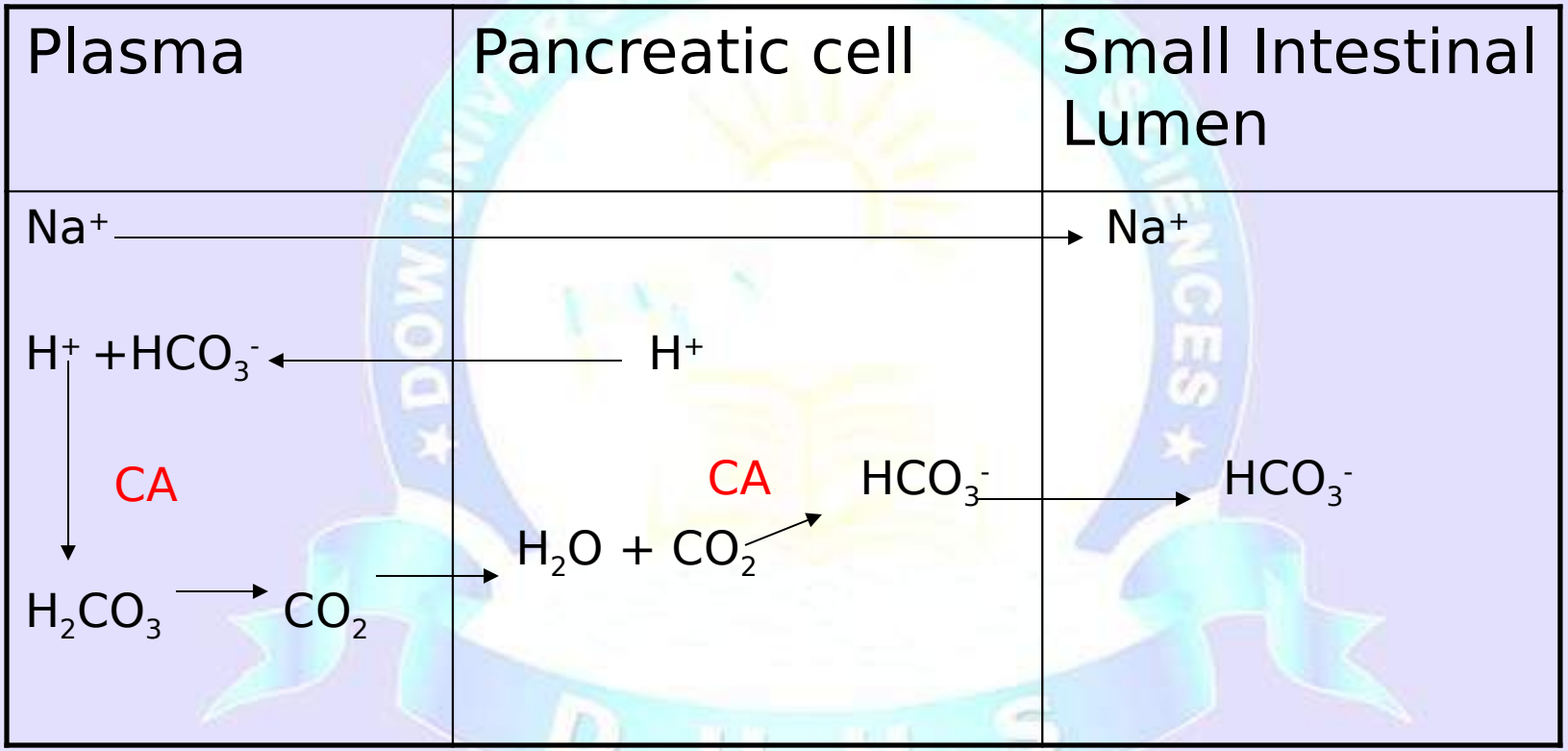
Transport of carbon dioxide working with hemoglobin



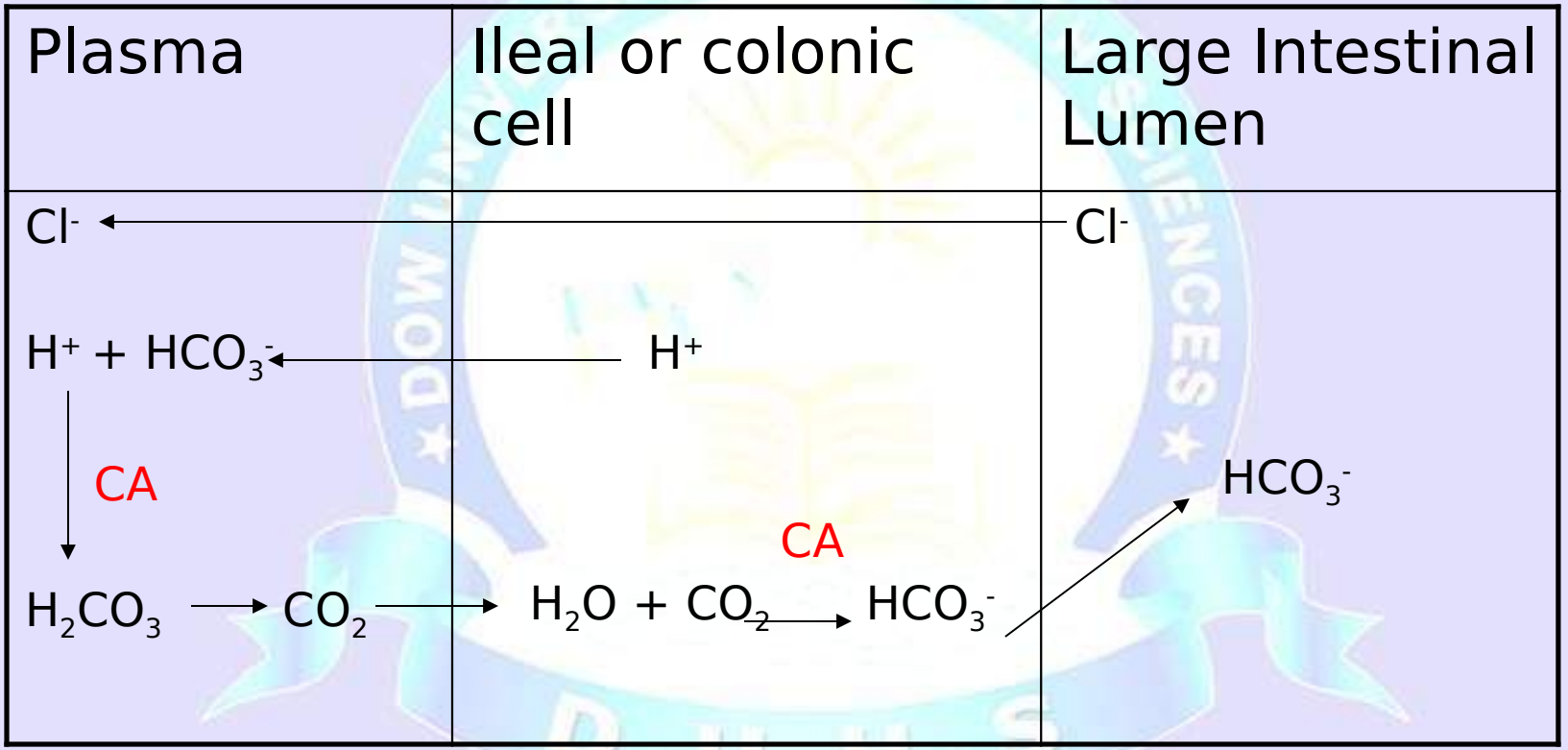
Working in Stomach



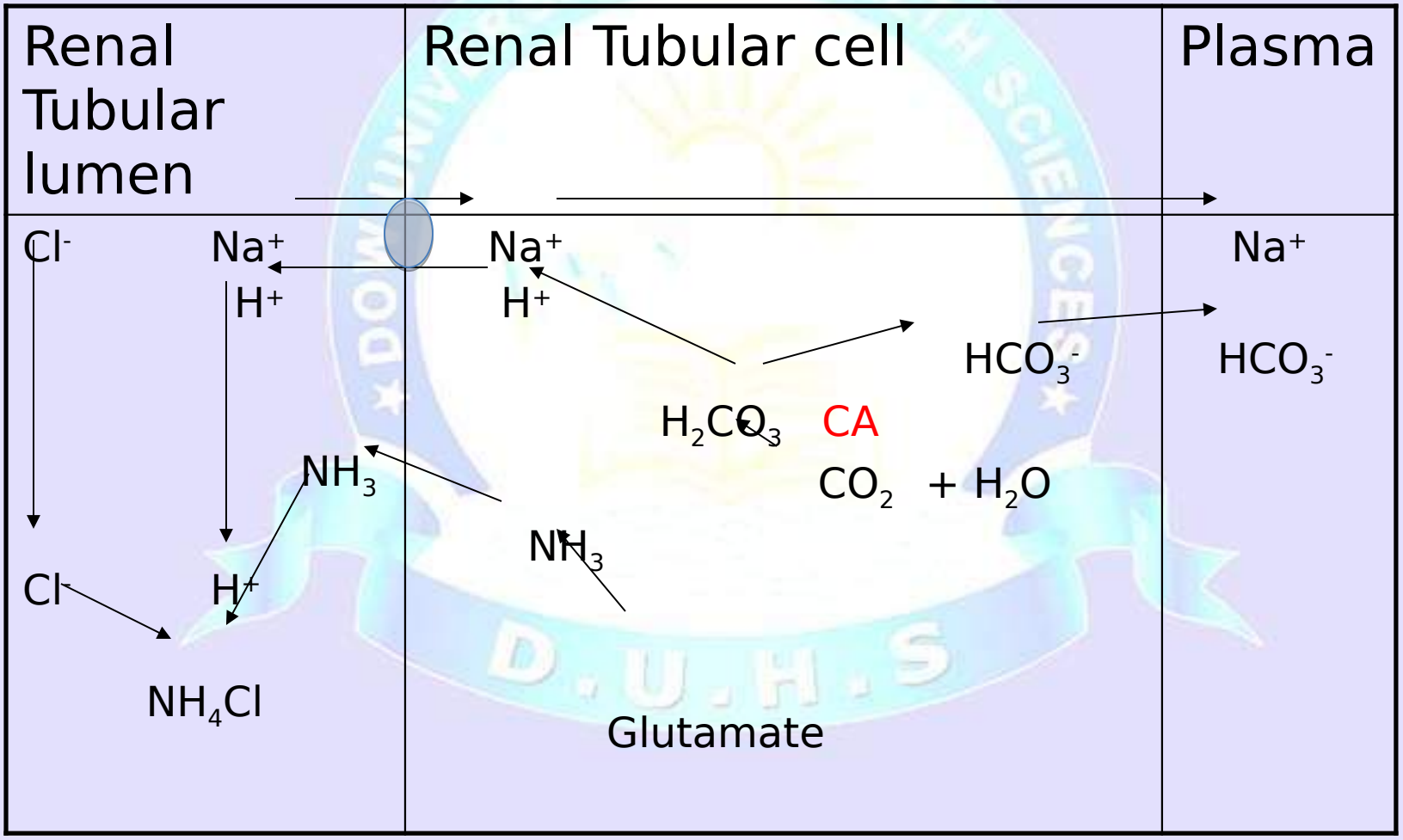
Working in Small Intestine



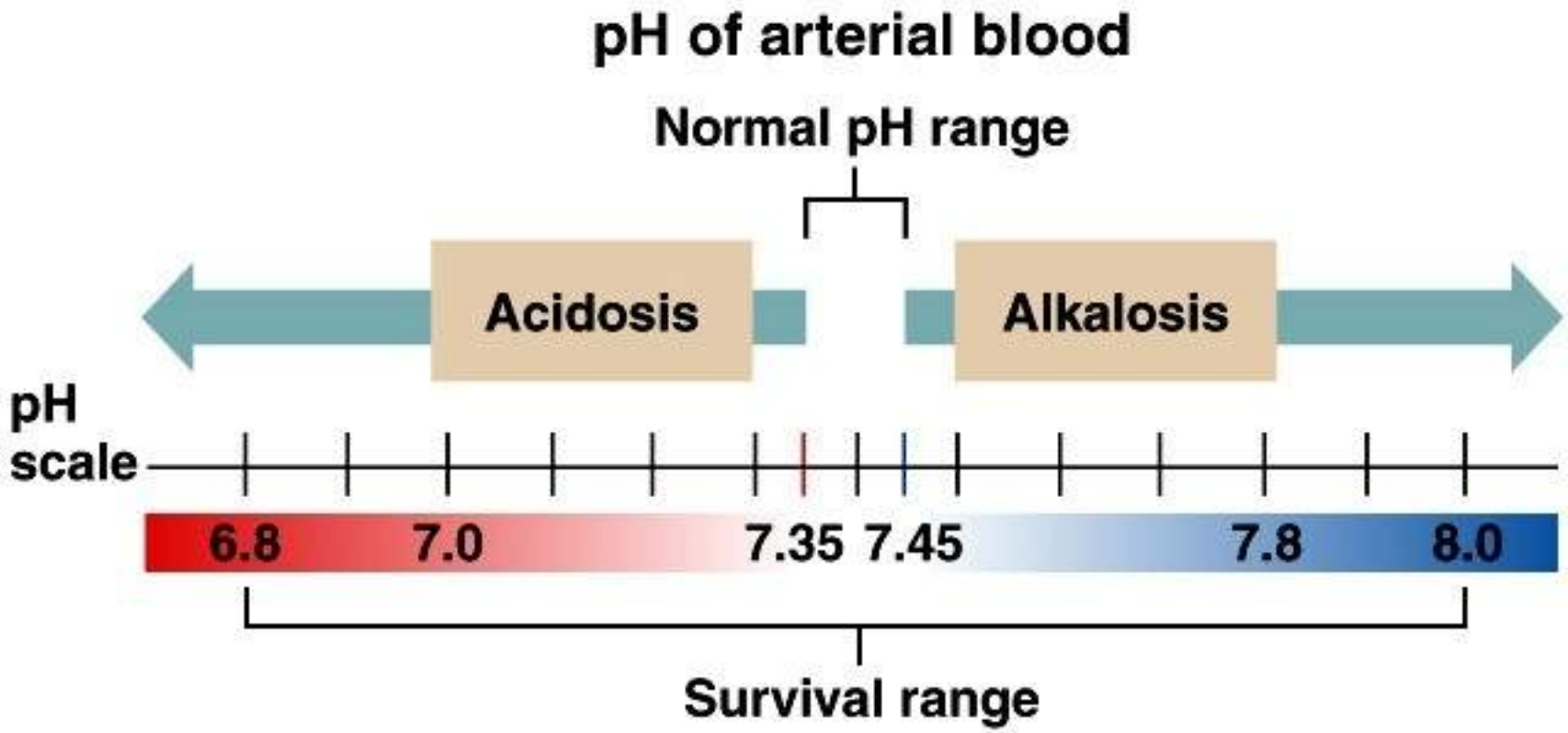
Working in large Intestine

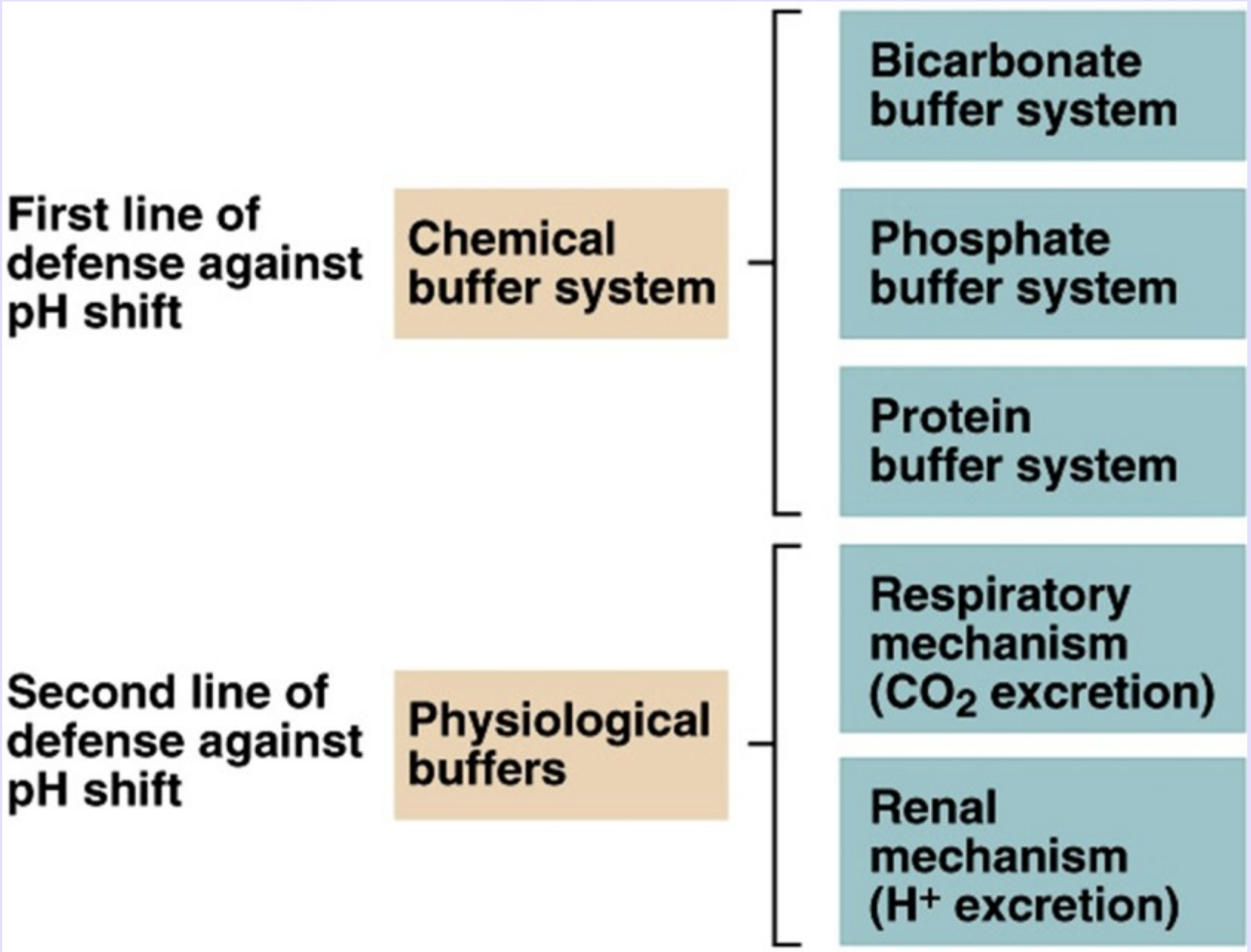


Working with renal buffers

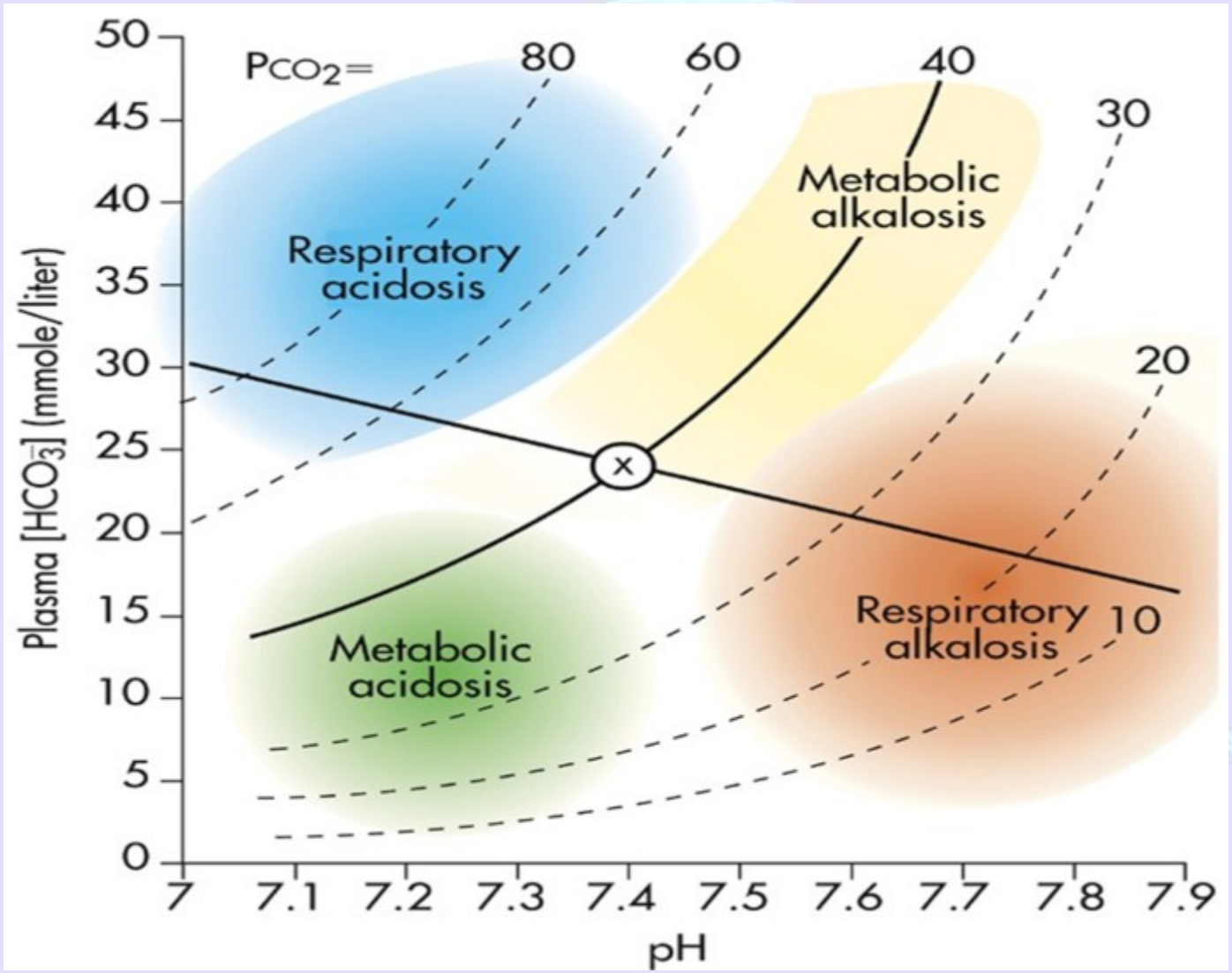


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First line of defense against pH shift
Chemical buffer system
Bicarbonate buffer system
Phosphate buffer system
Protein buffer system
Second line of defense against pH shift
Physiological buffers
Respiratory mechanism (CO_2 excretion)
Renal mechanism (H^+ excretion)

pH Derangements



REFRENCES

Harper chapter 1

THANK YOU